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MATRIC NO: _____
PRACTICUM: _____

Answer all questions.

- 1 Adam is a content creator for ABC company. He always makes live broadcast to promote a new product to his followers.

- (a) Name **two (2)** most suitable multimedia element that Adam can be used to promote his product on social media platform.

[2 marks]

Video, audio/video, text/video, graphic, animation

- (b) Describe **one (1)** purpose of using multimedia element mention in question 1(a).

[2 marks]

- Video-captures the audience's attention more effectively
- Video-enables real-time communication between the broadcaster and the audience
- video provides opportunities for audience interaction through comments, questions, and polls.
- Audio - allows broadcasters to convey information, share narratives, and communicate with the audience in a clear and expressive manner.
- Audio, including music, sound effects, and vocal intonations, can evoke emotions and enhance the overall mood of the broadcast
- facilitates two-way communication, enabling broadcasters to interact with their audience through live chats,
- Graphics are used to display important information, such as headlines, statistics, charts, and other data relevant to the content.
- build recognition and reinforces the broadcaster's identity.
- Text- display product name
- Animation - to make audience more intention

- (c) Identify **two (2)** common file formats mention in **1(b)** that Adam used on his social media platform.

[2 marks]

Mp3,mp4,wmv,jpg,gif,txt,ai,au,mpeg2,mov

- 2** A multimedia product is a product that combine different content forms such as text, audio, images, animation, or video into one single product. Multimedia products can be used for a variety of purposes, including education, entertainment, and marketing.

- (a) List **three (3)** processes that need to go through before a multimedia application is produced.

[3 marks]

Pre-production,production,post-production

- (b) State the process involved based on statement given below:

[4 marks]

Statement	Process
The first question we ask is “why” you want to develop a multimedia project?	Pre-production
The edited scripts are used to plan the identify location, performers, time schedules and budget, then the shoot is scheduled.	Production
The product is tested on multiple computers and monitors.	Post Production
Determine the basic of what types of equipment the audience have and what obstacles must be outcome.	Pre-production

- (c) Authoring tools enables users to create a digital course and publish it in select formats. There are **three (3)** types of authoring tools. Identify each type.

[3 marks]

Page base

Icon base

Time base

(d) Describe **two (2)** differentiate of this type of authoring tools.

[2 marks]

- elements and events are organized along a **timeline**.
- elements are organized **as pages of book or stack of cards**
- elements and events are organized **as objects in a structural framework or process**.

(e) HD media player or HDD media player (HDMP) is a consumer product that combines digital media player with a hard drive (HD) enclosure with all the hardware and software for playing audio, video and photos to a television.

- i) HD media player is one example of offline multimedia delivery. Give **two (2)** example of online multimedia delivery which can also work as HD media player.

[2 marks]

Cloud base,web base /

Youtube channel,Astro channel

- ii) Describe **one (1)** differentiate between offline and online multimedia delivery.

[2 marks]

Online	Offline
Can be changed and updated at any time by the developer.	Permanent and not changeable.
Require broadband internet connection	Can be accessed without internet

- 3 Many actions we do every day like use an ATM, shop online and many more generate data that needs to be stored and manage and used by others.

(a) Define database.

[2 marks]

A database is a collection of related data, organized in a manner that allows access, retrieval, and use of that data

(b) Give **one (1)** difference between file processing system and database system.

[2 marks]

File Processing System	Database System
• Data Redundancy – • Each department has its own files in a file processing system, thus the same fields are stored in multiple files	• Reduced Data Redundancy • Data is only entered one time, and stored in only one file

(c) Describe **one (1)** advantage of using database system.

[2 marks]

- Reduced Data Redundancy - Data is only entered one time, and stored in only one file
- Easier to update anytime and anywhere
- Easier to access anytime and anywhere

- 4 Database can manage large amount of data efficiently and enable information to sharing.

DeptCode	DeptName	NoOfLecturers
D01	Chemistry	32
D02	Computer Science	28
D03	Biology	29

- (a) Based on table above, state the most suitable primary key.

[1 mark]

DeptCode

- (b) State the data type for **DeptName**

[1 mark]

Text/short text/varchar/String

- (c) Describe **one (1)** function of data dictionary.

[2 marks]

- to perform validation checks to maintain the integrity of the data or
- allows users to specify a default value for a field
- enable to create relationship between entity correctly
- enable programmer to make update of fields easily

- 5 One of the major reasons business, home and other purchase computer is for Internet access. The internet is a widely used research tool, providing society with access to global information and instant communications.

- (a) Explain how the Global Positioning system (GPS) works?

[2 marks]

- A navigation system that consists of one or more earth-based receivers that accept and analyse signals sent by satellite in order to determine the receiver's geographic location.
- Uses satellites and a receiver to determine the exact geographic location of the receiver

- (b) State **two (2)** other usage of the network besides the one mentioned in 5 (a)

[2 marks]

- The Internet Application
- Telephone Service
- Television and Radio Broadcasting

- Monitoring System
- Multimedia Networking
- Videoconferencing, Collaborative Computing and Telecommuting
- Telemedicine

(c) State the type of wired transmission media based on usage given.

[4 marks]

Usage	Type of wired transmission media
Better security for signals during transmission because they are less susceptible to noise.	Coaxial Cable/ Fiber-Optic Cable
One of the more widely used transmission media for network cabling and telephone system.	Twisted-Pair Cable
Primarily used by cable TV companies to connect their satellite antenna facilities to customer homes and businesses.	Coaxial Cable
This type of transmission media is faster, more reliable and more expensive.	Fiber Optic

(d) State the type of wireless transmission media based on given characteristics.

[4 marks]

Characteristic	Type of wireless transmission media
Enables the transfer of data from one device to another device such as a mobile computer, mouse, printer and smartphone.	Infrared/ Cellular Radio
The type of transmission media in a form of broadcast radio that is used widely for mobile communications specially wireless modems and mobile phones.	Cellular Radio

Typically used a radio waves that provide a high-speed signal transmission over long distance.	Microwaves and Satellite
Application such as air navigation, television and radio broadcasting, whether forecasting, video conferencing, global positioning system and internet connection use this type of transmission media.	Microwaves and Satellite

(e) Identify the communication device based on the given function.

[6 marks]

Characteristic	Type of wireless communication device
A device that allows computers and devices to transfer data wirelessly among themselves or to a wired network using wireless technologies.	Wireless access point/ Wireless router
A central device that connects devices in a wired network but only send data to the intended recipient.	Switch
Used to connect two LANs together.	router
A communication device that enables a computer to communicate over analog networking media.	modem
A device that enables a computer or device that does not have built in networking capability to access a network.	Network adapter/ Network interface card(NIC)
A device that connects multiple computers or other routers together on a network.	switches

- 6 The web is worldwide resources of information. People use internet technology to find and explore various information that have no limit.

(a) Define search tools.

[2 marks]

- Search tools is a program to locate the information on the internet
- software programs designed to help users find information on the internet
- refer to software applications, programs, or online services designed to help users find information on the internet or within a specific database

(b) Explain the characteristic of both types of search tools mention in 6(a).

[4 marks]

Search Engine – A program to carry out web searches by allowing the user to enter keywords that are run against a database

Subject directory – a program to carry out web searchers that allowing user to browse through list of web sites by subject in categories and sub categories

(c) Identify the type of search strategy below:

[4 marks]

Situation	Search Strategies
Use 'OR' to broaden your search by looking for several terms in a source whether they appear together or not.	Boolean Operators
Use special symbol "*" which allow you to search simultaneously for several words	Keyword /Wildcard
Use quotation marks (" ") makes it possible to search for important words/phrases instead of individual keyword	Phrases

Use particular search characteristics such as URL or web site to limit the search result.	Field searches
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- (d) Identify the types of web activities based on the given description following:

[4 marks]

Description	Types of Web Activity
Digital content design to engage and entertain individuals through media and services that can be accessed online.	Online entertainment
It involve online transactions, electronics payments and the exchange of product and services without need physical interactions.	e-commerce
users can create and interact with content and other users on the internet.	social media
Prominent way for individuals to access up-to-date news content from a wide range of sources.	online news

- (e) Privacy issues is are fundamental aspect of data management in the digital area.

- i) Explain data privacy.

[2 marks]

- The relationship between the collection and dissemination of data, technology, the public expectation of privacy and the legal and political issues surrounding them
- Data privacy refers to the right of an individual to control or restrict the collection, use, and disclosure of their personal information

ii) Gives **two (2)** example how to protect our privacy data while do web activities.

[2 marks]

- use encryption for sensitive email
- always enable 2 steps of verification
- Avoid sharing personal information, such as your home address, phone number, or birthdate, on public websites or social media platforms.
- Do not share your credit card information or other financial information on websites unless you are absolutely sure they are secure.

(f) The goal of IoT is to enable any objects to communicate and interact with each other, as well as with humans to provide various services and capabilities.

Explain **one (1)** application of IoT in home appliance.

[2 marks]

- manage devices remotely at home such as starting washing machine at certain time
- preparing groceries shopping list
- Smart Thermostats:
 - adjust temperature settings based on occupancy, and be remotely controlled via smartphones
- Smart Lighting Systems:
 - allows for the creation of smart lighting systems that can be controlled remotely via mobile apps or voice commands
- Smart Washing Machines and Dryers:
 - enabled laundry appliances can be controlled remotely, allowing users to start, stop, or monitor the progress of washing and drying cycles using a mobile app.
- Automated Vacuum Cleaners:
 - Robotic vacuum cleaners equipped with IoT capabilities can be scheduled and controlled remotely
- Voice-Activated Appliances:
 - enables users to control various IoT-enabled devices using voice commands, creating a seamless and hands-free experience.
- Smart refrigerators can track food inventory and let you know when you're running low on items

END OF QUESTIONS.